observers. Virtual photons and virtual gravitons are the particle aspects of electromagnetic vacuum fluctuations and gravitational vacuum fluctuations, respectively. See also *wave/particle duality.*

warpage of spacetime. Same as *curvature of spacetime.*

wave. An oscillation in some field (for example, the electromagnetic field or space-time curvature) that propagates through spacetime.

waveform. A curve showing the details of the oscillations of a wave.

wavelength. The distance between the crests of a wave.

wave/particle duality. The fact that all waves sometimes behave like particles, and all particles sometimes behave like waves.

white-dwarf star. A star with roughly the circumference of the Earth but the mass of the Sun, which has exhausted all its nuclear fuel and is gradually cooling off. It supports itself against the squeeze of its own gravity by means of electron degeneracy pressure.

world line. The path of an object through spacetime or through a spacetime diagram.

wormhole. A “handle” in the topology of space, connecting two widely separated locations in our Universe.

X-rays. Electromagnetic waves with wavelength between that of ultraviolet radiation and gamma rays; see Figure P.2 on page 25.